Technical Research Report - Sensing System

Graduation internship

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Version 0.4, 08-09-2026

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1. Document history

Table 1. Document History

Version	Date	Status	Author	Description	Remarks
0.1	03-09-2026	Draft Version	Casey Stijlaart	Draft Version	Initial document layout, non-camera sensing techniques chapter
0.2	04-09-2026	Draft Version	Casey Stijlaart	Draft Version	draft version of sensing techniques
0.3	07-09-2026	Draft Version	Casey Stijlaart	Draft Version	Finalisation of non-camera sensing techniques + Initial draft of end-to-end latency chapter
0.4	08-09-2026	Draft Version	Casey Stijlaart	Draft Version	Finalised end-to-end latency + Initial draft of embedded compute platform chapter

2. Non-Camera Person Position & Height Sensing Techniques

2.1. Introduction

This chapter addresses the first technical research question defined in the Project Plan (Chapter 3.6, Research Questions), referred to as RQ1: "Which combination of non-camera sensors provides sufficient 3D spatial resolution to distinguish adult and child height categories within a room-scale interactive space?" Two further sub-questions are referenced by number: RQ2 (robust, low-latency position estimates under real-world noise) and RQ4 (extensibility to multi-person scenarios). RQ3 (on-device ML for height classification) is out of scope here.

Per the Project Plan (Chapter 3.3, Assignment), the target system estimates a person's real-time 3D position and height within the Holobox space without cameras, for the Virtual Human to act on. This chapter surveys candidate technologies against the criteria they must satisfy, compares current hardware and cost, and recommends a sensor choice with a staged de-risking plan rather than a final hardware order. A fully polished prototype is out of scope (Chapter 3.4, Scope and Preconditions). A general height estimate is treated as sufficient for now, and multi-person tracking is addressed only as an evaluation criterion, not a capability this stage builds.

2.2. Research Approach

This chapter follows the Library and Lab strategies of the DOT-framework identified for this research question in the Project Plan (Chapter 3.7, Research Methods). These are: available product analysis, literature study, document analysis, and bench experimentation.

2.2.1. Library: Literature Study

Candidate Sensor Technologies gathers and verifies peer-reviewed literature for each non-camera sensing family against the position/height requirement and, separately, against the multi-person active-speaker problem, in two separate passes. One sits in the radar/thermal/RF position-and-pose-estimation literature, the other closer to acoustic and vibrometry-based speaker localisation, so a single combined search would have missed sources specific to either.

2.2.2. Lab: Available Product Analysis

Alongside the literature, the same chapter narrows the literature-backed technology families down to specific, currently sourceable hardware modules, checked against live distributor listings (Digi-Key, Mouser, Adafruit, and others) rather than assumed pricing. An unverified cost estimate is not a usable basis for a hardware decision or the budget-approval process described in the Project Plan (Chapter 6.1, Finance and Risks). Literature backing and product availability are two different questions: a technique can be well-published without there being a specific, currently purchasable module that implements it at a reasonable price, or at all.

2.3. Evaluation Criteria

Before diving into available sensor modules, it is of great importance to understand the requirements they need to meet. Thus the comparison can be set out against explicit, justified requirements rather than an implicit sense of what "good" looks like. Each criterion below is traced to one of three sources: a technical sub-question in the Project Plan (Chapter 3.6, Research Questions), a key feature in the Assignment (Chapter 3.3), or a project constraint in Finance and Risks (Chapter 6.1). None are traced to a generic notion of sensor quality.

Table 2. Evaluation Criteria

Criterion	Description
Height-estimation capability	Must produce an actual height reading (alone or fused), not just (x, y) position, per RQ1's spatial-resolution requirement. A general estimate suffices for now.
Position accuracy and coverage	Position error and range must be usable at the Holobox's actual room-scale dimensions, since RQ1 covers (x, y) as well as height. A sensor that can't cover the space isn't usable regardless of height capability.
Real-time embedded feasibility	Interactive latency on an embedded platform (targeting <100 ms, hard ceiling ~1 s), as RQ2 requires low-latency embedded estimates. Thresholds follow Nielsen's (1993) response-time limits, building on Miller (1968).
Multi-person extensibility	Shouldn't fundamentally break if a second person enters the space later, since RQ4 and the Assignment both name multi-person as a future-facing requirement.
Data minimisation / privacy	Output can't identify an individual and captures only what's needed for position and height, required by the ethical/legal sub-questions given that visitors are children aged 8-12 (AVG/GDPR).
Literature and community backing	Prior published work must show the specific capability required, not just the technology in general, reducing the risk of committing project time to proving something never demonstrated at all.
Hardware cost	Cost of the sensor and any paired second sensor, checked against sourceable listings. A real constraint, since no project budget has yet been agreed with MindLabs.

These criteria are not weighted equally, and are not applied as a hard pass/fail filter. Height-estimation capability and data minimisation are treated as closer to hard requirements, since they follow directly from RQ1 and the ethical/legal sub-questions. The remaining criteria (sensor combination, cost, power/footprint) are weighed against each other in the chapter that follows instead. A technology that is merely more expensive or harder to pair with a second sensor is not disqualified the way one with no height precedent or an identifying output effectively is.

2.4. Candidate Sensor Technologies

This chapter surveys non-camera sensing technologies capable of device-free person detection, localisation, and (where the literature supports it) height estimation. These are evaluated against the technical sub-questions described in the Project Plan (Chapter 3.6, Research Questions): primarily sufficient 3D spatial resolution for height, embedded-platform feasibility, and the data-minimisation/privacy constraints that follow from the primary visitors being children aged 8-12. A general height estimate is sufficient for now; distinguishing adult and child height categories is a possible later refinement, not a hard requirement.

Each technology is covered once, in two parts: what the literature has actually shown it can do, and, for the technically viable candidates, how that holds up against the criteria set out earlier, specific hardware, cost, and pros/cons. The chapter then closes with a comparison summary, criteria matrix, and recommendation. Ngamakeur et al. (2020) frame this same comparison at a broader scale, structuring device-free sensing into a pipeline (Figure 1). It is used below to organise what each technology has actually been shown to do, rather than what it could in principle do.

Figure 1. Device-free Pipeline

2.4.1. Sensing Technologies

Where a technology has literature for both problems, its write-up splits into Position & Height Localisation and Speaker Identification subsections. Several technologies are covered for localisation only, and Laser Doppler Vibrometry and the Microphone Array for speaker identification only. None of the speaker-identification sources have been tested in this project's specific context (children's voices, a room-scale interactive installation), so they remain a lead for further work rather than a settled result.

2.4.1.1. Time-of-Flight & LiDAR

2D LiDAR measures range by timing a laser pulse's reflection while scanning a single horizontal plane, producing an (x, y) point cloud at the unit's mounted height. This is well suited to in-plane tracking, but a single fixed-height plane does not by itself yield a height measurement; that would need multiple units at different heights, a tilted scan, or a full 3D ToF/LiDAR unit.

Within its native (x, y) role, 2D LiDAR performs well: Bouazizi et al. (2024) fuse multiple units with a ConvLSTM model and show tolerance to furniture occlusion. For multi-person settings, Alam et al. (2021) document "crossover ambiguity" as 2D LiDAR's recurring multi-person failure mode when tracks pass close together, the concrete risk to plan around later. The representative module, the RPLIDAR A2M8, scans at 5-15 Hz (typically 10 Hz) per Slamtec's datasheet, the limit on full (x, y) frame availability regardless of its faster ~4000 samples/sec per-sample rate.

Pros

- Strong in-plane position accuracy and furniture-occlusion tolerance when multiple units are fused (Bouazizi et al.).
- Mature, widely available consumer/robotics hardware (RPLIDAR family) with broad community support.

Cons

- No height reading from a single scan plane; needs a second sensor or added mechanical complexity to get one.
- "Crossover ambiguity" is a documented multi-person failure mode when tracks pass close together (Alam et al.).
- 5-15 Hz scan rate limits how quickly a full (x, y) frame becomes available.

2.4.1.2. mmWave Radar

mmWave radar has an extensive body of research literature covering both position tracking and skeletal/joint-level pose estimation. Its point-cloud output is also inherently non-identifying (no recognisable imagery), aligning with the sub-questions on minimising personally identifiable data from child visitors. Available product analysis narrows this to a 60 GHz point-cloud radar module built around the TI IWR6843 chip, over 24 GHz presence sensors (binary presence flag only, no point cloud) and undocumented generic modules lacking reference software. Several IWR6843 evaluation-board variants exist at different price points and field-of-view characteristics; Module Selection compares them and picks one to carry forward.

Position & Height Localisation

For position tracking, Huang et al. (2021) combine clutter removal, clustering, and Kalman filtering for real-time multi-person tracking on a Raspberry Pi 4. Pegoraro and Rossi (2022) separately report position error down to roughly 7.6 cm. Pose-estimation literature is also relevant here: Zhu et al.'s ProbRadarM3F (2024) recovers 14 body keypoints from radar point clouds, including head and shoulder joints. This is backed by a camera-free radar dataset with joint ground truth (Su et al., 2024). This project’s own planned approach is simpler, though: just take the top and bottom point of a person’s point cloud and measure the gap (see End-to-End Latency Requirements, Compute Platform Implications). It’s cheaper and faster, but unlike the keypoint model, it isn’t backed by literature yet. The keypoint model stays the fallback if the simple version doesn’t hold up. On latency: reported tracking papers give broadly similar tens-of-milliseconds frame periods on different radar chips (15-20 fps). TI doesn’t publish a fixed maximum frame rate for the IWR6843 chip itself; the mmWave SDK exposes frame periodicity as an open, configurable value (2022), not a spec. TI’s own indoor people-tracking reference design for this chip (2020) runs at roughly 18 Hz (55ms), in the same range as the 10-20 Hz "typical demo" configurations noted elsewhere. See Evaluation Criteria for the thresholds this is checked against.

Speaker Identification

Whether the IWR6843 reports two people as distinct targets depends on angular and range resolution, not a blanket "multi-person" capability: roughly 1.7-2.9m of horizontal separation, or a ~1.7m height difference, is needed before angular resolution tells them apart. Range separation is far finer, ~4-5cm, from time-of-flight alone.

mmWave's own speaker-ID path is Doppler vibrometry: sensing larynx skin-surface displacement via a ~90-200Hz bandpass. Xu et al.'s WaveEar (2019) demonstrates this, but only at short range. No source shows it working room-wide at the scale this project needs, so it's a bonus capability here, not the primary mechanism. In this design it confirms whoever is close to the Holobox (~1-1.5m), doubling as a liveness/anti-spoof check: a played-back recording has no matching throat-vibration signature.

Room-wide speaker identification is delegated to a paired Microphone Array instead, steered by the radar's per-person tracked angles. For an illustrative 3×3m tracked zone with 2-3 people, ordinary spacing satisfies counting and height. At this scale, though, two people already approach the 1.7-2.9m threshold at the zone's diagonal (~4.2m), so ambiguous tracks are reachable with fewer people than a larger space would allow. That zone size is a planning illustration, not a confirmed Holobox spec.

Pros

- Extensive literature base: real-time multi-person tracking and keypoint-level pose/height estimation.
- Demonstrated real-time operation on a Raspberry Pi 4.
- Point-cloud output is inherently non-identifying, fitting the project's data-minimisation requirements.
- Also supports close-range (~1-1.5m) speaker confirmation and liveness/anti-spoof checking via Doppler vibrometry, without additional hardware, alongside the room-wide microphone array.

Cons

- Sensor cost is €160-185 for the IWR6843AOPEVM variant (see Module Selection).
- Height estimation still needs extra work: either the simple top/bottom-point calculation or a keypoint-extraction model. Neither is a single, settled method yet.
- Radar-only speaker identification is not demonstrated room-wide; a paired microphone array is needed for that (see below).

2.4.1.3. Wi-Fi CSI Sensing

Wi-Fi Channel State Information (CSI) sensing infers presence and movement from fluctuations in the wireless channel caused by a body moving through it, using only commodity Wi-Fi hardware. The gathered literature is consistent, though, that it trades away spatial resolution and stability: CSI is sensitive to environmental changes (furniture, multipath, other RF sources).

Position & Height Localisation

IndoTrack (2017) is the foundational device-free CSI tracking paper in this space. More recent work narrows toward multi-person settings, e.g. Shen et al.'s time-selective RNN for busy-room detection (2023). Reported update rates vary by an order of magnitude across otherwise comparable commodity setups, from 10 Hz (Shen et al., capped by their router) to a reported 100 Hz (IndoTrack, unconfirmed against the primary paper). That spread is an additional latency risk on top of the lack of height output.

Speaker Identification

Wi-Fi CSI reflections from mouth/lip movement during speech are detectable using only commodity Wi-Fi: Wang et al.'s WiHear (2014) uses directional antennas to sense multiple people’s talk simultaneously via MIMO.

Pros

- Effectively free: reuses existing Wi-Fi infrastructure.
- Multi-person and multi-room literature exists (Time-Selective RNN).
- Device-free output with no recognisable imagery.
- The same commodity hardware also has a literature precedent for detecting mouth-movement reflections for speaker identification (WiHear).

Cons

- No published height-estimation precedent at all, the single biggest gap for this project’s needs.
- Sensitive to environmental changes (furniture, multipath, other RF sources), the primary limit on stability.
- No demonstrated real-time embedded deployment in the sources reviewed.

2.4.1.4. Ultrasonic Ranging

Active ultrasonic ranging measures distance via time-difference-of-arrival (TDOA) or time-of-flight on emitted pulses, sidestepping the "room full of talking people" interference problem a passive-listening approach would face.

Segers et al. (2015) track up to four transmitters at 1-2cm accuracy on an FPGA using frequency-shift-keyed ultrasound. Segers et al. do not report an update rate or latency figure, so latency here is an open question rather than a documented one.

Pros

- Sidesteps the "room full of talking people" interference problem passive listening would face.
- Demonstrated real-time operation on embedded hardware (an FPGA, per Segers et al.).
- Reasonable literature base for ranging and positioning.

Cons

- No gathered source addresses height or elevation.
- Full coverage needs multiple synchronised nodes, reintroducing multi-sensor complexity.
- Custom-built hardware, roughly €200-400 (rough estimate, not a single purchasable SKU).

2.4.1.5. Infrared / Thermal

Low-resolution thermal/thermopile arrays measure a grid of per-pixel temperature readings: a person appears as a warm blob. The same low pixel count that limits resolution and range doubles as a privacy strength, too coarse to resolve recognisable imagery.

Naser et al. (2021) extend thermal-array distance estimation to height using a neural network, the most directly relevant source, and should be read first. Shubha and Shastrimath (2022) show 99% occupancy accuracy on a Grid-EYE/AMG8833 array on a Raspberry Pi. Klir et al. (2025) extend the same sensor class to multi-person localisation. Naser et al.'s sensor samples at 0.5-64 Hz; the AMG8833 breakout identified for procurement is coarser, only 10 Hz or 1 Hz.

Hardware cost is around 2 × €40-55, confirmed at €49.60 (incl. VAT) via Kiwi Electronics (NL); the I2C interface is straightforward. That price is for the AMG8833's narrow ~60° FoV, enough only for a small room. Room-scale coverage without mounting overhead, not an option for the Holobox, instead means the wider 110° MLX90640 variant, €75.01 (incl. VAT) at Kiwi Electronics (NL). Paired with a room-scale primary sensor (Combined Sensor Approaches), two narrow-FoV AMG8833 units (~€100 total) split coverage instead of one pricier wide-FoV part.

Pros

- Height estimation is demonstrated directly from sensor data, not just a proxy (Naser et al.).
- Low sensor-only cost (€40-55 narrow-FoV; €70-75 for the wide-FoV variant needed for room-scale coverage).
- Demonstrated real-time embedded operation on a Raspberry Pi.
- Low pixel count doubles as a privacy strength: output cannot resolve recognisable imagery.

Cons

- Effective range constrains it to smaller rooms unless the wider-FoV variant is used.
- Lower spatial resolution overall, which needs checking against the Holobox's actual dimensions.

2.4.1.6. Capacitive & Floor-based Sensing

Capacitive / Electric-Field Sensing

Two device-free variants exist: proximity-plate sensing (measuring distance to a body via a change in capacitance) and floor-embedded capacitive electrodes (requiring a person to stand on the sensed area). Bian et al.'s survey (2024) covers the design space broadly, and Faulkner et al.'s CapLoc (2020) is a floor-embedded mat evaluated in the EvAAL indoor-localisation competition. Range is the main constraint: plate sensors top out around 0.3-1m, so whole-room coverage needs a floor grid or many nodes. The signal is also inherently presence/proximity rather than spatial, so it is not naturally suited to a height reading at all.

Floor-based (Pressure) Sensing

Floor-based sensing (pressure, piezoresistive, or capacitive-floor mats) is inherently a 2D floor-plane measurement, but because each footstep is a spatially discrete event, it naturally disambiguates multiple people. Gatto et al. (2015) use a 64x64 pressure-switch array per tile, and Ibara et al. (2013) show gait-based individual identification from stride length and direction.

Pros

- Floor-based variants trivially solve "how many people, where."
- Solid literature base for both sub-variants.
- Inherently non-identifying (presence/proximity or pressure, not imagery).

Cons

- No height signal at all from either sub-variant.
- Capacitive sensing has short range, needing a dense grid for whole-room coverage.
- Floor-based sensing requires physical installation under the existing floor; cost can run into the thousands of euros.

2.4.1.7. Laser Doppler Vibrometry

Speaker identification only; not evaluated for position/height localisation.

A laser Doppler vibrometer (LDV) measures skin-surface vibration at a distance via Doppler shift, picking up throat vibration during speech even in noisy conditions. Avargel and Cohen (2011) combine an LDV sensor aimed at the larynx with a standard microphone to enhance speech quality under transient noise.

2.4.1.8. Microphone Array (Passive Acoustic)

Speaker identification only; not evaluated for position/height localisation.

Berghi and Jackson (2023) localise an active talker from multichannel audio alone using a CRNN, no camera or radar needed. This project doesn’t need that open-ended version, though: the radar pipeline (mmWave Radar) already tracks each person’s position.

Room-wide Speaker Identification (Primary)

Instead of blind direction-of-arrival estimation, the array steers a few listening beams at the radar's tracked angles and compares acoustic energy across them to attribute speech. This is considerably more reliable than a blind approach, but dependent on radar tracking already working. An ML refinement of that beam comparison, rather than the blind CRNN approach, is worth exploring too, though not yet evaluated for feasibility here. This makes the microphone array, not the radar's own close-range Doppler vibrometry, the primary path to room-wide speaker identification, covering the room the radar's ~1-1.5m range can't reach.

Open Risk: Array Range vs. Tracked-Zone Coverage

Off-the-shelf arrays like the Seeed ReSpeaker are rated for ~3m. For the illustrative 3×3m tracked zone used elsewhere, an edge mount's worst case (~4.2m, the zone's diagonal) exceeds that; a central mount's (~2.1m) doesn't. Both the zone size and mounting position are undecided, so this stays open. A higher-gain array or a central mount are the two directions to check before hardware is committed to.

2.4.2. Combined Sensor Approaches

The individual technologies above are also weighed as sensor combinations, where pairing two of them addresses a gap neither has alone.

2.4.2.1. 2D LiDAR + Thermal Fusion

As with the other thermal-fusion options above, the thermal side here is two AMG8833 units (~€100 total). This splits room coverage between them rather than relying on one pricier wide-FoV MLX90640 or a single narrow-FoV unit that falls short on its own. The representative LiDAR module itself, the RPLIDAR A2M8, also now shows as discontinued/out of stock at major distributors (e.g. Seeed Studio), an added sourcing risk on top of the concerns below.

Pros

- 2D LiDAR contributes strong in-plane position accuracy and occlusion tolerance (Bouazizi et al.).
- Thermal contributes the same direct height-estimation precedent used elsewhere in this chapter (Naser et al.).

Cons

- Sensor-only cost of €350-390, driven up by needing two thermal units.
- Combining two independent position-sensing streams from scratch introduces synchronisation and fusion complexity flagged as a disproportionate implementation risk for this project's timeline, as described in the Project Plan (Chapter 4.3, Timeline).
- The representative RPLIDAR A2M8 module is discontinued at major distributors, an unresolved sourcing risk specific to this combination.
- Does not reuse an already-working single-sensor pipeline; both position-sensing streams are new to this combination.

2.4.3. Comparison Summary

Table 3. Candidate Sensor Comparison Summary

Technology	Representative Module	Approx. Hardware Cost (sensor only)	Height Data in Literature	Documented Latency / Update Rate
mmWave Radar	TI IWR6843AOPEVM (see Module Selection for the variant comparison)	€174,92-€248.23	Yes, keypoint-based	10-20Hz as base, but configurable to higher rates.
Infrared/Thermal Array	AMG8833 breakout, Kiwi Electronics (NL)	€48.39	Yes, direct	1 Hz or 10 Hz
2D LiDAR + Thermal (sensor fusion)	RPLIDAR A2M8 + 2x AMG8833, Kiwi Electronics (NL)	€385.69 + 2x €48.39	Partial, via fusion	5-15 Hz
Wi-Fi CSI Sensing	None. It reuses existing commodity Wi-Fi hardware	Approx. €0 (existing infrastructure)	None found	10-100 Hz
Ultrasonic Ranging	Custom transducer array (e.g. SPM0404UD5 MEMS elements, per Segers et al. (2015))	€200-400 (rough estimate for custom-built hardware, not a single purchasable SKU)	None found	Not reported
Capacitive & Floor-based Sensing	Custom-built capacitive/pressure floor mat. There is no off-the-shelf module	€80-2000+, depending on variant (rough estimate for custom-built hardware)	None found	Not reported

Figures above are sensor/module hardware only. All candidates other than Wi-Fi CSI also need an embedded compute platform to run the sensing pipeline. That cost is excluded here since it is a shared requirement across candidates rather than something that distinguishes one sensor choice from another. Latency figures are the sensor's own documented output rate, not a measured end-to-end pipeline latency; see the caveat already noted in Evaluation Criteria.

Table 4. Criteria Comparison Matrix

Technology	Height Est.	Embedded Real-Time	Multi-Person	Fusion-Ready	Privacy	Literature	Price Range (sensor only)
mmWave Radar	✔	✔	✔	✔	✔	✔	€160-250
Infrared/Thermal Array	✔	✔	✔	✔	✔	✔	€80-110
2D LiDAR + Thermal Fusion	✔	□	✔	✔	✔	✔	€350-390
Wi-Fi CSI Sensing	□	□	✔	□	✔	✔	~€0 (existing infrastructure)
Ultrasonic Ranging	□	✔	□	□	✔	✔	€200-400
Capacitive & Floor-based Sensing	□	□	✔	□	✔	✔	€80-2000+

2.4.4. Recommended Approach

On the criteria set out earlier, mmWave radar comes out ahead. It’s the only approach with keypoint-level pose literature, and it has the strongest literature base of any candidate reviewed. The infrared/thermal array is cheaper and has a direct height-estimation precedent, but mmWave is still the stronger technical choice. This is a recommendation of the modality; the specific IWR6843 variant is addressed in Module Selection. Height itself is only a general estimate for now, adult/child categorisation is a possible later refinement, not something this recommendation locks in.

For the RQ4 speaker-identification problem, radar alone only reaches ~1-1.5m around the Holobox, via Doppler vibrometry, which doubles as a liveness/anti-spoof check. Room-wide speaker identification is instead handled by a paired Microphone Array, steered by the angles radar already tracks for each person, rather than solving open-ended source localisation on its own.

mmWave is by far the most expensive candidate reviewed. The company mentor has since approved that cost, though, so hardware ordering can proceed (Module Selection). Implementation risk is managed by building the pipeline one capability at a time: position, then height, then multi-person clustering, then speaker detection, each on an already-working previous phase. The microphone array gets added once radar tracking works, since beam-steering depends on it, and only once its own hardware-range risk (see Microphone Array) is resolved.

2.5. Module Selection

Candidate Sensor Technologies recommends mmWave radar as the sensing modality, built around the TI IWR6843 chip. The chip ships on several evaluation-board variants at different price points and field-of-view characteristics, given below as azimuth (angular coverage side-to-side) / elevation (angular coverage top-to-bottom, i.e. floor-to-head reach). This chapter compares those variants directly and picks one. The retrieved product listings are current as of 07-09-2026 and may vary when checking on a different date.

- IWR6843ISK: the standard long-range antenna plug-in module, €248,23 at current Mouser NL listings, with a 120° azimuth / a narrower 30° elevation field of view tuned for range over coverage. That trade-off works against a room-scale height-estimation use case rather than for it.
- IWR6843ISK-ODS: a wide-field-of-view (120° azimuth / 120° elevation) antenna variant, etched on a Rogers substrate. TI markets it specifically for overhead/ceiling-mounted people counting and tracking (~12 m range), €205,44 at current Mouser.
- IWR6843AOPEVM/AWR6843AOPEVM: an antenna-on-package (AoP) variant, with the antenna integrated directly into the chip package rather than etched onto a separate PCB, €174,92 at current Mouser NL/AWR6483AOPEVM. Matches the ISK-ODS's wide 120° azimuth / 120° elevation field of view in a smaller (~65×60mm).

Table 5. mmWave EVM Variant Comparison

Feature	IWR6843ISK	IWR6843ISK-ODS	IWR6843AOPEVM/AWR6843AOPEVM
Price (€)	248,23	205,44	174,92
Antenna	60-64GHz	60-64GHz	60-64GHz
Rx/Tx	4/3	4/3	4/3
Azimuth Angular Resolution (deg)	15	29	29
Azimuth FOV (deg)	120	120	120
Elevation Angular Resolution (deg)	58	29	29
Elevation FOV (deg)	30	120	120
Max Distance for child/adult (+/-m)	50/60	40/48	41/49

NOTE The maximum distance for the TI modules is given by the TI mmWave Sensing Estimator, configured for the short-range default; it is therefore theoretical, not a measured figure.

2.5.1. Recommendation

Of the three variants in Table 5, the IWR6843AOPEVM (equivalently, AWR6843AOPEVM) is recommended. Elevation, not azimuth or range, is the deciding factor. Elevation FOV and resolution determine how much of a person's vertical extent falls inside the beam, the precondition for the keypoint-based height estimate. Every variant already covers the room's short range with margin to spare. Azimuth isn't a deciding factor either. All three variants share roughly the same 29° figure anyway (bar IWR6843ISK's finer 15°). The fine angular discrimination room-wide speaker identification needs is handled by a paired microphone array instead (Candidate Sensor Technologies, Microphone Array), not the radar's own azimuth resolution.

IWR6843ISK is ruled out: its 30° elevation FOV is by far the narrowest of the three, its elevation resolution the coarsest, and it is also the most expensive at €248,23. IWR6843ISK-ODS matches AOPEVM's elevation FOV and resolution exactly, but costs 17% more and its antenna is built for ceiling mounting, an assumption the Holobox has not committed to. AOPEVM reaches the same elevation performance in a smaller, mounting-agnostic form factor at the lowest price of the three, which is why it is carried forward into the phased mmWave implementation in the Conclusion.

NOTE Update: the company mentor has approved the cost of the IWR6843AOPEVM/AWR6843AOPEVM, so this module can now be ordered.

2.6. Conclusion

Candidate Sensor Technologies and Module Selection recommend mmWave radar, specifically the IWR6843AOPEVM: the only approach with keypoint-level pose literature for height, non-identifying output, and the strongest literature base reviewed, at the lowest price of the elevation-capable variants compared.

Implementation risk is managed by building the pipeline in phases, position, then height, then multi-person clustering, then speaker detection, each on an already-working previous phase. mmWave is the most expensive candidate reviewed, though, so mentor approval of that cost is a precondition. Sensor fusion (mmWave or 2D LiDAR with thermal) wasn't adopted as the starting point. mmWave alone already has a plausible path to height, no fusion combination is tested together in the literature, and thermal needs two units to match mmWave's coverage. Fusion stays a documented fallback, not a precondition for starting.

Speaker identification (RQ4) splits across two sensors. mmWave's own Doppler-vibrometry path only reaches ~1-1.5m around the Holobox, a bonus that doubles as a liveness check, not a room-wide solution. Room-wide speaker-ID is handled by a paired microphone array instead, steered by angles the radar already tracks (Candidate Sensor Technologies, Microphone Array), added once radar tracking works. Open risk: off-the-shelf arrays like the Seeed ReSpeaker are rated for ~3m. Whether that's enough depends on mounting position relative to the tracked zone (illustrative 3×3m example): an edge mount's worst case (~4.2m) exceeds that rating, a central mount's (~2.1m) doesn't. Worth a higher-gain array or committing to central mounting before hardware is ordered.

3. End-to-End Latency Requirements for Natural Virtual Human Reaction

3.1. Introduction

This chapter addresses the User/Human Interaction sub-question defined in the Project Plan (Chapter 3.6, Research Questions), referred to as RQ8. "What position and height estimation latency is required for the Virtual Human's reaction (e.g. kneeling down for a child) to be perceived by visitors as natural and immediate, rather than delayed or robotic?"

Per the Project Plan (Chapter 3.3, Assignment), the Virtual Human must react to a visitor's detected position and height, downstream of the sensing pipeline covered in Non-Camera Person Position & Height Sensing Techniques. That reaction needs to read as immediate and natural, not delayed or robotic. Building or verifying that reaction isn't this project's job, though: Finished Products lists only the embedded sensing system as the prototype deliverable. The Virtual Human's own rendering is MindLabs' existing Holobox/Unreal Engine system. So this chapter sets two things. First, a full-pipeline target for RQ8 (sensing all the way to the visible reaction), derived from general HCI response-time literature and one telecom standard, since no literature specific to this reaction exists. Second, a narrower NFR that this project actually builds and can bench-test, sensing through to exposing the estimate to the pipeline. It breaks that budget down using the sensor already recommended there (Section 2.4.1.2, Section 2.5). As with the sensing research, this chapter proposes a target and a verification plan, not a measured result. The bench and field measurement in Requirements and Verification hasn't run yet. That's the next step, not something this desk-research stage claims to have already confirmed.

3.2. Research Approach

This chapter follows the Library and Field strategies of the DOT-framework identified in the Project Plan (Chapter 3.7, Research Methods): literature study and document analysis to derive and budget the latency target.

3.2.1. Library: Literature Study

The "feels natural" requirement is grounded in general response-time literature rather than a number picked from intuition. Miller (1968), Nielsen (1993), and Shneiderman & Plaisant (2010) establish the standing 0.1s/1s/10s interactive-system response-time thresholds. ITU-T G.114 (2003) gives a comparable, independently-derived threshold (150ms one-way) for real-time telecom traffic. A targeted search for literature on position/height-triggered avatar-reaction latency specifically, as opposed to general system responsiveness, did not surface a source that could be verified against its own claim. Perceptual Latency Thresholds is explicit about this gap, and derives a project-specific target from the general literature instead of attributing a number to a source that does not actually make that claim.

3.2.2. Field: Document Analysis

Latency Budget decomposes the sensing-to-reaction pipeline into per-stage budgets (sensor acquisition, fusion, inference, rendering, optional transmission) by analysing existing documentation rather than running new experiments. The sensor acquisition figures reuse the recommended mmWave module's own datasheet and SDK configuration options (Section 2.4.1.2, Section 2.5); the remaining stages are estimated from published figures for comparable embedded pipelines, since this project has not yet built and measured them.

3.3. Perceptual Latency Thresholds

3.3.1. General Interactive-System Thresholds

Miller (1968) establishes, and Nielsen (1993) and Shneiderman & Plaisant (2010) both restate, a three-tier response-time structure for interactive systems:

- ≤ **0.1s**: perceived as instantaneous; the user feels direct control over the system.
- **0.1s - 1s**: a noticeable but acceptable delay; the user's flow of thought stays uninterrupted.
- **1s - 10s**: noticeable enough that the system should give explicit feedback (e.g. a progress indicator), or the user's attention starts to drift.
- > **10s**: the user disengages from the task entirely.

These thresholds describe general system responsiveness (a button press, a page load), not a reciprocal physical reaction between two agents specifically. The Holobox's Virtual Human reacting to a visitor's detected position and height (e.g. kneeling for a child) is closer to the latter, so the next section narrows the target rather than applying the general 1s "acceptable" threshold as-is.

3.3.2. A Comparable Real-Time Threshold: ITU-T G.114

ITU-T Recommendation G.114 (2003), the telecom standard for one-way transmission time in real-time voice/video traffic, independently arrives at a similar boundary from a different domain. Below 150ms one-way delay is rated acceptable for most real-time applications, 150-400ms is acceptable but shows increasing impairment, and above 400ms is not acceptable. G.114 measures one-way, mouth-to-ear delay for telephony and video calls; it does not cover embodied position/height-reaction latency. It's cited here only as a second, independently-derived data point that a sub-150ms target sits inside a boundary already treated as "real-time acceptable" elsewhere, not as a source that makes a claim about virtual humans.

3.3.3. Project Target: Position/Height-Reaction Latency

Neither source above speaks to this project's specific reaction, position and height detection driving a Virtual Human behaviour change, so this project sets its own target rather than borrow one that doesn't fit. That target is < 150ms end-to-end (sensing to visible reaction), with < 100ms as a stretch target. The reasoning: a reciprocal physical response, the Virtual Human kneeling for a detected child, sits inside Miller/Nielsen/Shneiderman & Plaisant's "acceptable" band (0.1-1s) only at its very low end. The visitor isn't issuing a command and waiting for a result the way those thresholds assume. They're perceiving a continuous, two-way physical interaction instead, closer to G.114's mouth-to-ear immediacy than to a page-load delay. 150ms is therefore adopted as the hard ceiling for the full pipeline, matching G.114's "acceptable" boundary, with 100ms as the stretch target matching the general literature's "instantaneous" threshold. This project only builds and can verify part of that pipeline, though, so Requirements and Verification turns this full-pipeline target into a narrower, actually-testable NFR for the part this project owns.

3.4. Latency Budget

3.4.1. Pipeline Breakdown

The pipeline this project actually builds and can verify is sensing → fusion → inference → output (the estimate exposed to the Unreal Engine pipeline). It's broken into independent per-stage budgets below, so a measured overrun can later be pinned on a specific stage, not just "the system" as a whole. This is a broader budget than the <100ms embedded-feasibility target already set for the sensing stage alone (Table 2). That figure only covers sensing and position-estimation; this one covers everything up to the point this project's own deliverable ends. Sensor acquisition reuses the already-recommended module (Section 2.4.1.2, Section 2.5), not an assumed frame rate. The remaining stages, fusion, inference, output, are estimated from typical figures for comparable embedded pipelines, since this project hasn't built and measured them yet.

Component	Budget	Notes
Sensor acquisition (mmWave radar)	50-100 ms	10-20Hz base configuration, consistent with the "typical demo" figures in Section 2.4.1.2 and with TI's own indoor people-tracking reference design for this chip (2020), which achieves ~18Hz (55ms). The mmWave SDK allows configuring higher rates; see Sensor Sampling Rate below.
Sensor fusion (Kalman/EKF over multiple frames)	0-20 ms	Lightweight relative to inference; can run in parallel with the next frame's capture.
Inference (height-category classification)	1-5 ms	Theoretical estimate for a simple geometric approach: take the highest and lowest point in the person's radar point cloud, compare against an adult/child threshold. No trained model. Not yet measured on target hardware, and adult/child classification itself isn't a confirmed VH requirement yet (see below and Non-Camera Person Position & Height Sensing Techniques).
Output exposed to the pipeline (only if sensing and the pipeline consumer run on separate machines)	0 ms local / 50-100 ms networked	Negligible on one local machine; rises materially over Wi-Fi/Ethernet if split across machines, per Non-Camera Person Position & Height Sensing Techniques.
Total, this project's own scope	51-125 ms	Sum of the ranges above, local only. This is the part this project actually builds and can bench-test.

Downstream of that, MindLabs' existing Holobox / Unreal Engine system renders the Virtual Human's actual reaction, one animation frame updating skeletal targets, typically 16-33ms at 30-60 FPS. That's not something this project builds, controls, or can bench-test; Finished Products lists only the embedded sensing system as the prototype deliverable. It's mentioned here purely as context. Added to this project's own 51-125ms, it lands around 67-158ms, which is what the <150ms full-pipeline target (Perceptual Latency Thresholds) is actually measuring against.

3.4.2. Jitter Margin

Adding headroom for scheduling jitter and frame misalignment, a planning margin pending the bench measurement in Requirements and Verification, not a measured figure:

```
Base budget, this project's own scope (mid-point of each range above): ~88 ms
Jitter margin:                                                          +20 ms
Resulting estimate:                                                     ~108 ms
```

~108 ms is the planning target for NFR-Latency-01, this project's own scope, sensing through to exposing the estimate. That leaves roughly 42ms of the full 150ms budget for the downstream Virtual Human rendering this project doesn't build, close to the 16-33ms typically estimated for one render frame. That margin rests on an unvalidated assumption, though. If the simple top/bottom-point approach isn't accurate enough on real, noisy point-cloud data, and a heavier model is needed after all, inference could grow back toward 30-50ms and eat most of that margin. The bench measurement in Requirements and Verification is where this actually gets tested, not assumed.

3.4.3. Sensor Sampling Rate

TI doesn't publish a fixed maximum frame rate for the IWR6843AOP. The mmWave SDK treats frame periodicity as an open, configurable float value (2022). The only real constraints: active-chirp duty cycle stays ≤50% of the frame period, and there's enough time for the UART transfer. No documented ceiling. So this chapter plans around a 10-20Hz base rate for the already-selected IWR6843AOPEVM (Section 2.5), the same "typical demo" range from Section 2.4.1.2, rather than picking one number out of thin air. TI's own indoor people-tracking reference design for this chip (2020) backs that up: a 288-chirp, 1.78GHz-sweep profile, tuned for 8.4cm range resolution over 6m, runs at 55ms (~18Hz) with a 46% duty cycle, within the SDK's ≤50% limit. That's a real, published example, not just a demo default. Since frame periodicity is open, running faster is possible and would ease the budget further. But TI hasn't published a faster worked example for this use case. A leaner chirp profile, trading range/velocity resolution for speed, is worth exploring, but that's for the bench measurement in Requirements and Verification to check, not something to assume here.

3.4.4. Compute Platform Implications

Height-category classification has two options: a simple top/bottom-point gap check (no trained model, ~1-5ms), or the keypoint-extraction model from Non-Camera Person Position & Height Sensing Techniques (ProbRadarM3F, 2024, ~30-50ms). Neither figure is measured or literature-backed for this task, but ML is likely overkill for a gap this simple. The simple approach is primary. The keypoint model is the fallback if it isn't accurate enough against real, noisy point clouds (occlusion, missed points, clustered people), untested either way. Even the fallback's speed is unproven here: Huang et al. (2021) show tracking on a Raspberry Pi 4, not classification.

This also assumes the VH's behaviour depends on adult/child category at all, not confirmed yet (Non-Camera Person Position & Height Sensing Techniques). And if inference runs on a paired host, that host also runs the microphone array's beam-steering (Candidate Sensor Technologies, Microphone Array), a compute-contention risk the bench measurement should check for.

3.5. Requirements and Verification

3.5.1. Non-Functional Requirements

NFR-Latency-01	This project's own pipeline (radar frame arrival to output exposed to the Unreal Engine pipeline) shall take < 110ms (p50) and < 150ms (p95), nominal local-processing conditions.
Rationale	The full-pipeline target is <150ms (p50), sensing to visible VH reaction. But VH rendering is MindLabs' existing system, not this project's deliverable (Finished Products). 110ms is this project's own slice, leaving ~40ms for that downstream rendering.
Measurement	Timestamp radar frame arrival, fusion output, inference result, output exposed. Report median/p95 delta, 10-minute window, N ≥ 20 trials.
Dependencies	Radar frame rate in the 10-20Hz base range (Section 2.4.1.2, 2020). Inference ≤ 5ms (simple approach) or ≤ 50ms (ML fallback); both theoretical. No competing background process during measurement.

3.5.2. Verification Strategy

This NFR isn't confirmed yet; it needs the height-classification pipeline to exist first. Once it does:

1. Inject a test stimulus at a known time (e.g. a person of known height stepping into the tracked zone on cue).
2. Log timestamps: radar frame capture, fusion output, inference result, output exposed.
3. Compute the delta from stimulus to that exposed output, per trial (not to the VH's own reaction, that's out of scope).
4. Repeat for N ≥ 20 trials over 10 minutes, to capture run-to-run variance, not one lucky run.
5. If inference runs on a paired host, repeat with the microphone array's beam-steering workload running concurrently, to check for compute contention.
6. Report median/p95 latency against NFR-Latency-01.

Local-only first (no network hop), to isolate this from Unreal Engine transmission latency. Real-user field testing and sprint-review feedback, per the Project Plan's research methods, come after, once local measurement confirms the pipeline is in-budget.

3.6. Conclusion

RQ8 sets a full-pipeline target of < 150ms (p50) / < 200ms (p95), sensing to visible VH reaction, from general HCI thresholds and ITU-T G.114. But that full reaction isn't this project's deliverable; only the embedded sensing system is (Finished Products). VH rendering is MindLabs' existing Holobox/Unreal Engine system. So NFR-Latency-01 covers a narrower, measurable slice: sensing through to exposing the estimate, < 110ms (p50) / < 150ms (p95).

That narrower budget has some room, on paper, under an unvalidated assumption. No documented max frame rate exists for the IWR6843AOPEVM, so this chapter plans around TI's 10-20Hz base rate (Section 2.4.1.2, 2020, ~18Hz), and assumes height classification is a simple top/bottom-point calculation, not a trained model. That leaves ~108ms against the 110ms ceiling, not much room. That margin depends on the simple approach holding up, or falling back to the heavier keypoint model, itself unproven for this task (Huang et al., 2021, show tracking on a Raspberry Pi 4, not classification). The bench measurement in Requirements and Verification is where this gets confirmed.

As with Non-Camera Person Position & Height Sensing Techniques, this chapter is a target and a verification plan, not a finished result. Every figure, frame rate, inference cost, fusion time, margin, is theoretical, pending validation once a working pipeline exists.

4. Embedded Compute Platform Selection

4.1. Introduction

This chapter addresses RQ2 and RQ3 from the Project Plan (Chapter 3.6, Research Questions). RQ2 covers implementing the sensor on an embedded platform for robust, low-latency position estimates; RQ3 covers deploying a lightweight classification model on that platform. Non-Camera Person Position & Height Sensing Techniques already fixes the sensor, the TI IWR6843AOPEVM, whose onboard DSP and Cortex-R4F run TI's tracking firmware and produce a point-cloud stream. Its only interfaces are UART-over-USB and a GPIO header, though, with no Ethernet or WiFi of its own, so it cannot reach the Holobox pipeline on its own. A host board is therefore needed purely to bridge that gap: receive the stream, run classification, and put the result on the network. Candidate Sensor Technologies and End-to-End Latency Requirements both already assume this board exists without naming or costing one.

Running classification on the machine already driving the Holobox's Unreal Engine pipeline was considered and rejected. NFR-Latency-01 would prefer it (0ms local vs. 50-100ms networked, Latency Budget), but the Project Plan describes the deliverable as its own embedded hardware and software prototype (Assignment, Finished Products), not a service bolted onto MindLabs' existing show PC.

This chapter picks that host board and checks it against the latency budget and cost constraints already established. It covers only whether the board has spare compute for the microphone array's beam-steering workload once added (Candidate Sensor Technologies, Microphone Array), not the array hardware itself. It also does not train or validate the classification model; that is RQ3's model-accuracy question, addressed separately once a working pipeline exists.

4.2. Research Approach

This chapter follows the Lab and Library strategies the Project Plan assigns to RQ2 and RQ3 (Chapter 3.7, Research Methods): literature review, available product analysis, and prototyping/benchmarking. The in-situ and model-training parts of those rows come later, once a board is chosen and a pipeline exists to benchmark.

4.2.1. Library: Literature Study

The search here is narrower than Non-Camera Person Position & Height Sensing Techniques: rather than surveying sensing modalities, it looks for published or documented cases that pair this project's already-selected radar family (the TI IWR6843 series) with a specific class of host board. The goal is to check whether the interface and compute split this project plans to use has actually been demonstrated, not just assumed.

4.2.2. Lab: Available Product Analysis

Candidate boards are narrowed to specific, currently sourceable hardware, checked against live distributor listings rather than assumed pricing, for the same reason as Non-Camera Person Position & Height Sensing Techniques. An unverified cost estimate is not a usable basis for the budget-approval process in the Project Plan (Chapter 6.1, Finance and Risks). Literature precedent and product availability are again treated as separate questions: a board can be well-documented for general embedded use without a published case of it running this project's specific sensor pipeline.

4.3. Evaluation Criteria

As with Non-Camera Person Position & Height Sensing Techniques, candidate boards are compared against explicit criteria. Each criterion traces back to a research question, an already-established budget, or a project constraint, rather than a general sense of which board is "more powerful."

Table 6. Evaluation Criteria

Criterion	Description
Host interface compatibility	Must receive the already-selected IWR6843AOPEVM’s tracked point-cloud output without added interface hardware, since the sensor module is fixed (Module Selection).
Real-time classification headroom	Must run height classification inside the ~108ms per-project latency budget (Latency Budget), per RQ2’s low-latency requirement. This applies to both the primary simple threshold approach (~1-5ms) and, if it becomes necessary, the ProbRadarM3F keypoint fallback (~30-50ms).
Concurrent workload headroom	Must have spare compute if there is necesaiity to add a microphone array.
Output path to the Holobox pipeline	Needs a network interface so exposing the estimate does not force it onto the same machine as Unreal Engine; local-only adds 0ms, networked adds 50-100ms (NFR-Latency-01).
Literature and community precedent	Must show a documented case pairing this exact radar family with the board class, not just general embedded capability, for the same reason as Evaluation Criteria in Non-Camera Person Position & Height Sensing Techniques.
Hardware cost	Cost of the board, checked against sourceable listings; still a real constraint, since no project budget has yet been agreed with MindLabs (Finance and Risks).
Power and footprint	Must suit mounting near or inside the Holobox installation: passive or small-fan cooling, not a rack-mounted or actively loud unit.
Availability at ISSD	Whether the board can be borrowed directly from Fontys' own ISSD hardware loan desk. Since MindLabs is itself a Fontys initiative, a board available there can be obtained without waiting on the mentor budget approval that every purchased candidate still needs (Finance and Risks).

Host interface compatibility and real-time classification headroom are treated as closer to hard requirements. A board that cannot receive the sensor’s own output, or cannot fit inside the latency budget already committed to in Latency Budget, is not usable regardless of cost or literature backing. Availability at ISSD is also treated as close to decisive rather than just another weighed factor: it is the only criterion that can remove the pending-budget blocker entirely, rather than just keeping the ask small. The remaining criteria are weighed against each other in the chapter that follows.

4.4. Candidate Embedded Platforms

The IWR6843AOPEVM's own onboard DSP and Cortex-R4F already run TI's people-tracking firmware and produce a tracked point-cloud stream (2020, 2022). The host board this chapter picks does not run the mmWave SDK itself. Its job is narrower: consume that stream, run the classification stage, and expose the result to the Holobox pipeline. Barral et al. (2024) document exactly this split on the same radar family. TI's onboard firmware handles point-cloud generation and a first-pass extended-Kalman-filter track, while their paired host runs the downstream filtering, clustering, and output stack.

4.4.1. Bare-Metal MCU

Finance and Risks' own risk table already names an MCU-to-Linux-SBC fallback direction, implying the project could in principle start at the MCU tier. A representative part, the ESP32-S3, is cheap (a few euros) and has built-in WiFi. No gathered source pairs an MCU-class part with this radar family for point-cloud consumption, though. Running the sensor's output parser, a classification stage, and later the microphone array's beam-steering concurrently is also a level of scheduling and memory management an MCU without a real OS is not well suited to for this project's timeline. This tier is ruled out as a starting point rather than attempted first and abandoned. The project starts directly at the Linux single-board-computer (SBC) tier the risk table already anticipates.

4.4.2. Raspberry Pi 4 Model B (4GB/8GB)

A quad-core Cortex-A72 @ 1.5GHz, Gigabit Ethernet, dual-band WiFi, and USB 3.0/2.0, sold in 2GB (€60.49), 4GB (€108.89), and 8GB (€157.29) RAM variants at current Kiwi Electronics (NL) listings, all sharing the same CPU and interfaces.

The 4GB variant is carried forward: the design skips Barral et al.'s ROS-based stack, and classification runs in the IWR's own onboard firmware rather than on the host (Introduction). The host's job therefore reduces to relaying a small classified result over UART, running the microphone array's beam-steering, and bridging to the network, none of which approaches a 4GB ceiling. The one scenario needing more memory, the ProbRadarM3F fallback's PyTorch runtime, is also a CPU-throughput problem, so escalating to the Raspberry Pi 5 (Recommended Approach) addresses both rather than buying 8GB now.

This is the only candidate with a literature precedent for the exact combination this project plans to use. Barral et al. (2024) pair an IWR6843ISK, the same radar family as the already-selected IWR6843AOPEVM, with a Raspberry Pi 4 over USB, running TI's onboard tracking firmware on the radar and a downstream ROS-based clustering/tracking/output stack on the Pi. Huang et al. (2021) separately show real-time multi-person mmWave tracking running on the same board.

Barral et al. also report a concrete open risk worth carrying into the bench-testing stage: bandwidth constraints extracting measurements over the USB interface at higher data rates, unconfirmed against this project's own (as yet unmeasured) point-cloud rate.

Pros

- Only candidate with a literature precedent for this exact radar-family-plus-host combination (Barral et al.).
- Separate real-time multi-person tracking precedent on the same board (Huang et al.).
- Cheapest board reviewed (€108.89 incl. VAT) and the only one available to borrow from Fontys' ISSD, keeping the initial ask small while budget approval is pending (Finance and Risks).
- Standard GPIO/UART-over-USB interface and mainstream Linux ecosystem, no added interface hardware needed.
- 4GB comfortably covers the host's reduced job, since classification runs on the IWR itself.

Cons

- No onboard GPU/NPU acceleration. ProbRadarM3F can still run on CPU, but whether it stays inside the latency budget on this board is unproven; it would also need the board to fall back to ingesting the raw point cloud rather than the lean classified stream.
- Barral et al.'s reported USB bandwidth constraint is an open risk, unconfirmed against this project's own data rate, and would only actually matter if that same fallback is triggered.

4.4.3. Raspberry Pi 5

A quad-core, out-of-order Cortex-A76 @ 2.4GHz with 8GB RAM, Gigabit Ethernet with PoE+ support, WiFi, USB 3.0, PCIe 2.0, and a 40-pin GPIO/UART header, €192.38 incl. VAT at current Kiwi Electronics (NL) listings.

No gathered source pairs this specific board with the IWR6843 family. It shares the same GPIO/UART interface and Debian-based Linux stack as the wider Raspberry Pi family, though, so integrating this project's sensor-parsing and classification software needs no new interface work. Its out-of-order CPU cores and doubled RAM give real headroom for the ProbRadarM3F keypoint fallback (~30-50ms estimate) and for running the microphone array's beam-steering concurrently, though neither is benchmarked on this exact pipeline yet. PCIe 2.0 and USB 3.0 also leave room for a future accelerator add-on, if CPU-only inference on this board still proves insufficient.

Pros

- Meaningfully more CPU and memory headroom than the primary classification approach needs on its own, a margin against the ProbRadarM3F fallback or concurrent microphone-array beam-steering.
- Same GPIO/UART interface and Linux ecosystem as the wider Raspberry Pi family, so no new interface work is needed.
- PCIe and USB 3.0 leave room for a future accelerator add-on if CPU-only inference proves insufficient.

Cons

- No gathered source pairs this specific board with the IWR6843 family.
- €192.38 incl. VAT is a real expense given no project budget is approved yet (Finance and Risks).
- Not available at Fontys' ISSD; adopting it now would mean waiting on mentor budget approval before it can even be obtained.

4.4.4. NVIDIA Jetson Orin Nano Super Developer Kit

A 6-core Cortex-A78AE @ 1.7GHz paired with a 1024-core Ampere GPU and 32 tensor cores (up to 67 TOPS INT8), 8GB LPDDR5, 7-25W power envelope. No EUR-market listing was found at a major distributor. Converting Electrokit's (SE) listed 5,329 SEK incl. VAT gives approximately €465, consistent with Aetherix's (NL) directly listed €468.47, so €465-470 is used as the working estimate.

No gathered source pairs this board, or any Jetson model, with the IWR6843 family or with the ProbRadarM3F keypoint model specifically. The case for it is inferred, not demonstrated: ProbRadarM3F (2024) is a PyTorch/CUDA model, and Jetson's CUDA and TensorRT stack is the standard embedded deployment target for that model class in general. That is a reasonable inference from how such models are normally deployed, not a result specific to this project's use case.

Pros

- Only candidate with GPU/tensor-core acceleration, the class of hardware the ProbRadarM3F keypoint fallback would most plausibly need if the simple threshold approach underperforms.
- Same GPIO-class headers and USB/networking interfaces as the SBC candidates, so the sensor interface itself is no harder to integrate.

Cons

- No literature or documented case pairs it with this project's specific sensor or classification model.
- At ~€465-470 incl. VAT, this is a substantial per-unit cost against Finance and Risks' unapproved budget.
- A 7-25W power envelope and active cooling requirement, a consideration if the board needs to sit inside or close to the Holobox housing.
- Justified only if the simple threshold classification approach turns out to be insufficient; committing to it now would be pre-paying for a capability not yet shown to be needed.
- Not available at Fontys' ISSD; adopting it now would mean waiting on mentor budget approval before it can even be obtained.

4.4.5. Comparison Summary

Table 7. Candidate Embedded Platform Comparison Summary

Board	Compute	Cost (VAT)	Precedent	Power	ISSD	Fallback Headroom
Raspberry Pi 4 Model B (4GB/8GB)	Quad-core Cortex-A72 @ 1.5GHz	€108.89/€149.99	Yes, direct (Barral et al.; Huang et al.)	~5-8W	Yes	None beyond primary; ML fallback unproven on CPU
Raspberry Pi 5 (4GB/8GB)	Quad-core Cortex-A76 @ 2.4GHz	€158.07/€192.38	No (same ecosystem as Pi 4)	~8-12W	No	More CPU/RAM; ML fallback ~30-50ms est., no GPU
Jetson Orin Nano Super Dev Kit	6-core Cortex-A78AE + Ampere GPU	€539	None found	7-25W	No	Full: GPU/tensor cores handle ML fallback directly

Figures are board-only manufacturer specifications, not measured on this project’s own pipeline; cost uses the Raspberry Pi 4’s 4GB price throughout. Precedent, cost, and ISSD availability decide the starting platform: the Pi 4 is cheapest, has the only literature precedent, and is the only candidate available to borrow. The Pi 5 costs about 77% more and the Jetson roughly 5x, and both would need to be purchased. Power draw and fallback headroom are secondary. The Jetson’s GPU/tensor cores give it the most headroom, at 3-5x either Pi’s power draw, but that only matters if bench testing later triggers the ProbRadarM3F fallback or a concurrency problem the Pi 4 can’t absorb.

4.4.6. Recommended Approach

The Raspberry Pi 4 Model B (4GB) is recommended as the starting platform. It is the only candidate with a literature precedent for this exact radar-plus-host split (Barral et al.), reinforced by a separate real-time-tracking precedent on the same board (Huang et al.). It is also the cheapest board reviewed, and the only one available to borrow from Fontys' ISSD, avoiding the pending budget approval entirely. 4GB is sufficient since classification runs on the IWR itself, not the host; the host only relays a small classified result, runs the microphone array’s beam-steering, and bridges both to the network (Introduction).

This is a starting-point recommendation, not a final hardware order, and still needs mentor sign-off. Two risks carry into bench testing (Research Approach): Barral et al.'s reported USB bandwidth constraint, and whether the Pi 4 has spare CPU cycles once beam-steering runs alongside classification. Neither is a RAM problem, so paying for 8GB now would not address either. If either risk materialises, the Raspberry Pi 5 is the documented fallback: same interface and ecosystem, roughly 2.2-2.4x the compute headroom (Geekbench 6.2) for a 77% cost increase, and not available at ISSD. The Jetson Orin Nano Super remains a further fallback specifically for the ProbRadarM3F model, only if the primary classification approach proves insufficient. It is the most expensive and highest-power candidate reviewed, and like the Pi 5 it would need to be purchased rather than borrowed.

4.5. Conclusion

Candidate Embedded Platforms recommends the Raspberry Pi 4 Model B (4GB) as the host board that receives the IWR6843AOPEVM’s classified position/height result, runs the microphone array’s beam-steering, and bridges both to the Holobox pipeline. It is the only candidate with a literature precedent for this combination (Barral et al.), the cheapest option with any sensor-family precedent, and it fits the latency budget already committed to (Latency Budget). 4GB is sufficient since classification runs in the IWR’s own onboard firmware, not the host. It is also the only candidate available to borrow from Fontys' ISSD, avoiding the pending mentor budget approval (Finance and Risks).

This is a starting-point recommendation pending mentor sign-off, not a final hardware order. Two risks carry into bench testing: Barral et al.'s reported USB bandwidth constraint, and whether the board has spare compute once beam-steering runs alongside classification. The Raspberry Pi 5 and Jetson Orin Nano Super remain documented fallbacks for either risk, neither adopted now since neither risk is confirmed and neither board is available at ISSD. Every figure here is a manufacturer specification or literature precedent, not a measurement from this project’s own hardware.

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